EBU6304 Software Engineering Group Project

2025-26

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First report

1. Requirements

1.1 Fact-finding Techniques

1.1.1 Background Reading

Collect and sort out existing documents of the International College, including TA recruitment management regulations, TA work demand descriptions of each module per semester, historical recruitment Excel data, and feedback summaries from Module Organizers (MOs) and applicants in previous recruitments; at the same time, analyze similar TA recruitment systems of other domestic and foreign universities, conduct competitive research, and identify the advantages and disadvantages of existing systems (such as the convenience of resume upload and the rationality of position matching). It provides a high-level overview of the core components and interaction logic for system construction, laying a foundation for subsequent targeted research.

1.1.2 Interviewing

As the most widely used fact-finding technique, the team conducted one-on-one interviews, focus groups, and seminars targeting core stakeholders—Module Organizers (MOs), TA applicants (students), and college administrators:

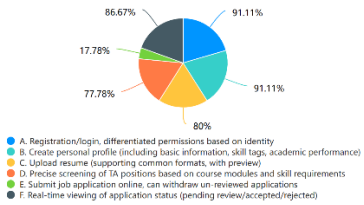
- **One-on-one interviews**: Interview 9 stakeholders including MOs from different academic modules, students who have participated in TA recruitment, and college administrators in academic management. Each interview lasts 30-40 minutes, focusing on the pain points of the current recruitment process (such as the difficulties MOs face in screening applicants and the challenges administrators encounter in workload statistics) and specific functional expectations for the new system;
- **Focus group seminar**: Organize 11 student applicants who have participated in TA recruitment to conduct a 90-minute seminar, discussing their needs in terms of resume upload convenience, position search accuracy, and real-time query of application status.
- **Seminar**: Gather multiple stakeholders including Module Organizers (MOs), college administrators, and TA applicants. Through cross-role interactive communication, clarify the business logic and data flow rules of the entire TA recruitment process, coordinate the demand differences and potential conflicts among different roles, and finally form a system function framework and business specifications that take into account the core demands of all parties.

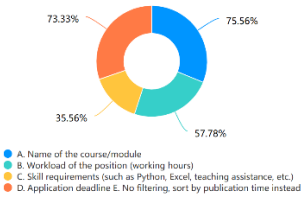
This technique achieves in-depth and personalized excavation of requirements, directly obtaining the real demands of different stakeholders.

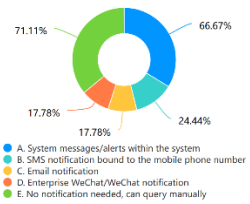
1.1.3 Questionnaires

To collect the needs of a large number of student applicants and MOs (exceeding the interview sample size), an online questionnaire was designed and distributed. The questionnaire includes 15 questions (single-choice, multiple-choice, and open-ended), covering dimensions such as the frequency of using the recruitment system, functional requirements, expected system response speed, and user interface experience. A total of 57 valid questionnaires were collected (45 from students and 12 from MOs). Statistical analysis of the questionnaire results was conducted by computer, efficiently collecting user input and supplementing the detailed needs of the general stakeholders that cannot be covered by interviews.

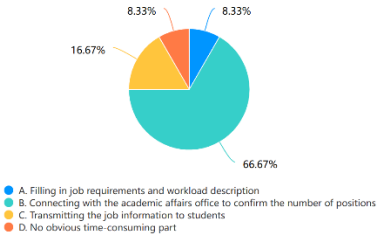
From the questionnaire results (as shown in the figures), it can be analyzed that student applicants highly value basic functions such as account permission management, personal profile setup, resume upload, and precise TA position screening; when filtering positions, the course/module name and application deadline are the most critical criteria; for notification methods, in-system messages/alerts are the first choice, with compatibility for multi-channel pushes such as WeChat, while only a small number of users prefer manual querying.

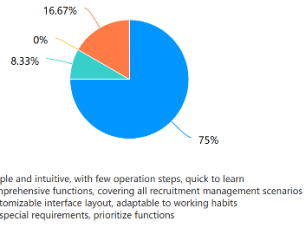


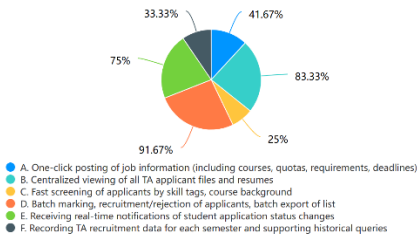


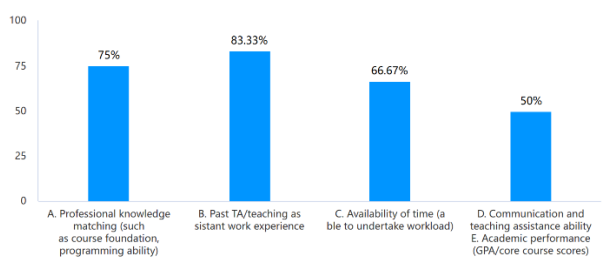


From the questionnaire results (as shown in the figures), it can be analyzed that Module Organizers (MOs) find confirming position quotas with the academic affairs office the most time-consuming step in job posting; when screening applicants, they prioritize past TA/teaching assistant experience, professional knowledge matching, and time availability; core system demands focus on batch operations, centralized resume viewing, and efficient applicant screening, with a strong preference for simple, intuitive, and easy-to-learn user interfaces.









1.2 Iteration Planning

Combined with the TA recruitment system project timeline and Agile Scrum methodology, the team divided the entire development process into 4 iteration cycles, each strictly aligned with the project evaluation nodes. All cycles follow the Scrum cycle: cycle planning → daily stand-up meeting → development and testing → sprint review → retrospective summary. The core objectives, deliverables, and time nodes of each cycle are highly consistent with the Product Backlog and project evaluation requirements (first evaluation on March 22, mid-term evaluation on April 12, final evaluation on May 24).

1.2.1 Core Cycle Rules

1. **Incremental delivery**: Each sprint delivers a runnable software version with testable core functions, and the final version integrates all sprint deliverables;
2. **Continuous feedback**: After each sprint, collect feedback from Teaching Assistants (TAs) and stakeholders, and adjust the Product Backlog and subsequent sprint plans accordingly;
3. **GitHub collaboration**: Assign independent development tasks to each team member, with each task corresponding to a dedicated branch, and regularly initiate pull requests to the main branch (to track individual contributions).

1.2.2 Detailed Sprint Plan

Sprint No.	Time Range	Core Objective	Core Deliverables	Alignment with Evaluation Nodes
1	March 22 - March 30	Complete the basic system construction and MVP core functions to meet the requirements of the first evaluation	NFR-01/02/03/04, US-01/02, TA-01 (File Management)	First evaluation (Product Backlog, Prototype, Brief Report)
2	April 4 - April 11	Complete the full TA-side business process and deliver runnable Software Version 1	TA-02/03/04 (Position Viewing/Application/ Status Tracking)	Mid-term evaluation preparation
3	April 12 - April 29	Complete the core MO-side functions and basic administrator supervision functions, and deliver runnable Software Version 2	MO-01/02/03, AD-01/02	Mid-term evaluation (Demo defense, GitHub contributions)
4	May 4 - May 24	Complete administrator workload management, AI auxiliary functions and system integration, and deliver the final product	AD-03/04, AI-01/02/03, System Testing and Optimization	Final evaluation (Final software, Demo video, Final report)

1.2.3 Daily Agile Practices

Daily stand-up meeting (15 minutes): Held online/offline every working day to track task progress, report blocking issues, and align daily work plans;

Sprint review: Held after each sprint to demonstrate the runnable software version to the course teaching assistant and collect feedback;

Sprint retrospective: Analyze the highlights and problems in the sprint work (such as task delays, poor communication) and put forward improvement measures for the next sprint;

GitHub management: The main branch is used for stable versions, each task corresponds to a feature branch, code review must be performed before merging, and commit information must be detailed (recording development history).

1.3 Prioritisation

The team adopted the MoSCoW prioritization method (aligned with EBU6304 Agile practice requirements) to prioritize all 20 user stories (including functional and non-functional requirements) in the Product Backlog. The prioritization follows three core principles:

1. Core needs of stakeholders: Prioritize solving the most painful functions in the current recruitment process;

2. Project constraint compliance: All mandatory technical constraints (such as local text file storage, Java Servlet/JSP development) are implemented in early sprints;

3. Feasibility and iterative delivery: Low-complexity, high-value functions are prioritized in early sprints, and complex AI functions are scheduled for the final sprint (as optional functions).

The definition, proportion, and sprint distribution of the four priority levels are as follows:

1.3.1 Must have (M)

Definition: Core requirements without which the system cannot realize basic TA recruitment functions, as well as mandatory technical constraints and non-functional requirements (NFR);

Proportion: 14 user stories (accounting for 70% of the total);

Core examples: NFR-01 (Local Storage), US-01 (User Registration), TA-01 (TA Profile Management), MO-01 (TA Vacancy Posting), AD-03 (Workload Report), NFR-04 (Error Prompt Usability).

1.3.2 Should have (S)

Definition: Important requirements that improve system efficiency and usability, non-core but of high value to stakeholders;

Proportion: 3 user stories (accounting for 15% of the total);

Core examples: AD-01 (Lobby Business Management), AD-02 (Administrator Dashboard), AD-04 (Workload Constraints).

1.3.3 Could have (C)

Definition: Value-added functions that are icing on the cake, with low development urgency and can be cut when time is limited;

Proportion: 2 user stories (accounting for 10% of the total);

Core examples: AI-01 (Identify Missing Skills), AI-03 (AI Skill Matching).

1.3.4 Want to have (W)

Definition: features serve only a limited group of users and do not drive the same amount of business value as the preceding items.

Proportion: 1 user stories (accounting for 5% of the total);

Core examples: AI-02 (AI Workload Balance).

Resolution of Requirement Conflicts

For identified requirement conflicts (such as the conflict between TA applicants' requirement for multi-format resume upload and the system's lightweight storage requirement), the team adopted the stakeholder trade-off method to resolve them:

1. Identify the core stakeholders corresponding to the conflicting requirements;
2. Evaluate the development cost and functional value of each solution;
3. Select the solution that meets the core requirements with the minimum development cost (such as only supporting PDF/DOCX resume formats and limiting the file size to 20MB).

All conflict resolution decisions are recorded in the Product Backlog and confirmed by all team members.

1.4 Estimation Methods

To accurately estimate the development workload of each user story and each sprint, Story Points (SP) are used as the unified estimation unit. The estimation process is transparent and involves all team members to ensure the accuracy and fairness of the results.

One Story Point (SP) is defined as: the workload required by a developer with basic Java Servlet/JSP development capabilities to complete a simple, independent user story. Complex user stories are estimated with higher story point values, and the standard SP scale is 0, 1, 2, 3, 5, 8, 13 (aligned with the Fibonacci sequence to reflect task complexity).

Team members independently estimate the SP value of each user story based on task complexity, technical difficulty, and code volume; form a preliminary SP estimation table for the Product Backlog. Based on the consensus SP values of all user stories, combined with the prioritization results and sprint timeboxes, the total workload is allocated to each sprint. A 10% buffer workload is reserved for each sprint to deal with unexpected issues (such as bug fixes, requirement adjustments).

1.4.1 Summary of Estimation Results

Total project workload: 167 Story Points (SP);

Workload distribution of each sprint (aligned with iteration planning):

Sprint 1: 44SP (accounting for 25.6% of the total workload) — Basic system construction + MVP core functions;

Sprint 2: 29SP (accounting for 39.5% of the total workload) — Full TA-side business process;

Sprint 3: 47SP (accounting for 19.8% of the total workload) — Core MO-side functions + basic administrator functions;

Sprint 4: 47SP (accounting for 15.1% of the total workload) — Administrator workload management + AI functions;

Resource allocation: 6 team members (2 front-end, 2 back-end, 1 AI algorithm, 1 testing), with an average workload of about 27.8 SP per person (equally distributed according to personal skills).

Estimation Validity and Adjustment: The estimation results are dynamically adjustable: if user stories not completed in a sprint are moved to subsequent sprints, they are re-prioritized and the SP value is re-estimated; after each sprint, velocity analysis (actual completed SP vs estimated SP) is conducted to optimize the estimation method for subsequent sprints; all estimation adjustments are recorded in the Product Backlog and GitHub project wiki to ensure traceability.

2. Supporting material

Include supporting materials, such as interview questions, survey results, and customer feedback etc (if applicable).

2.1 Interview Results

2.1.1 Interview Results of Module Organizers (MO)

MO Interview Result 1

Q: How do you currently post TA position information? About how long does the whole process take?

A: Currently, I first post a notice in the college teachers' group, then sort out the detailed position information separately and send it to the study committee members of each class to forward. The whole process takes about 1.5 hours, mainly spent on formatting the information, forwarding through multiple channels, and then replying to students' questions one by one.

Q: What aspects do you mainly focus on when screening TA applicants? What inconveniences do you feel when using Excel for screening now?

A: When screening, I mainly focus on these points: the major must correspond to the module I teach, basic Excel skills are required, it is better to have TA or teaching assistant experience, and the time must be flexible enough to cooperate with the module's class and consultation hours. Screening with Excel is too troublesome. It is impossible to screen multiple conditions at the same time. For example, if I want to find students who "meet the major requirements + have TA experience", I have to screen repeatedly. Moreover, I cannot mark which ones have been viewed and which ones need re-interview, so it is easy to miss or recheck.

Q: What key information do you hope the new system will include when posting positions?

A: I hope it can include these key information: module name, number of TAs needed, weekly working hours (such as class and consultation hours), specific working time, necessary skills (like Excel, PPT and professional-related software), requirements for grade and major, and salary standard.

Q: Do you need batch operation or marking functions when viewing and screening applicants' resumes? If yes, in what situations do you usually use them?

A: Yes. For example, after screening out students who meet the basic conditions, I want to batch mark them as "to be re-interviewed", and those who do not meet the conditions as "eliminated", so I don't have to click line by line, which saves time. In addition, when I encounter outstanding and key applicants, I can also mark them for follow-up later.

Q: Do you need the system to automatically remind you of unprocessed applications? How often do you hope to be reminded and in what way?

A: Yes, otherwise it is easy to forget to process them. I hope to be reminded once a day at 5 pm, using system internal messages and SMS reminders, so as to ensure that important applications are not missed.

Q: Have you ever encountered a situation where the applicant's skills do not match the position during previous recruitments? How do you hope the system will help you solve this problem?

A: It often happens. For example, some students apply for TA positions in professional-related modules, but they are not familiar with the core knowledge points of the module or cannot use the software we need. I hope that when students submit their applications, the system can automatically compare the position requirements with the skills they fill in and pop up a matching prompt. In addition, I also want to set some core skill thresholds, so that students who do not meet them cannot submit applications, which can save a lot of screening time.

MO Interview Result 2

Q: How do you currently post TA position information? About how long does the whole process take?

A: I usually post it in the notice column of the college official website, and then send private messages to several interested students. The whole process takes about 1 hour, mainly spent on editing the official website notice, adjusting the format, and replying to students' private messages.

Q: What aspects do you mainly focus on when screening TA applicants? What inconveniences do you have when using Excel for screening now?

A: I mainly focus on students' core course scores, communication skills (whether they can clearly answer students' questions), and sense of responsibility. For example, those who have done part-time jobs or teaching assistants before are reliable, can work stably for a long time, and will not resign halfway. When using Excel for screening, it is impossible to quickly compare the information of multiple students, only line by line. The screened data cannot be exported quickly, and manual copy and paste are required, which is very inefficient. In addition, screening opinions cannot be recorded, so there is no unified record for reference when communicating screening results with other MOs later.

Q: What key information do you hope the new system will include when posting positions?

A: I hope it can include these: module number, specific work of TA (such as class assistance, homework correction, consultation, etc.), how to calculate workload, working location (online/offline), required skills (such as specific software, language ability), working period, and application deadline.

Q: Do you need batch operation or marking functions when viewing and screening applicants' resumes? In what specific situations do you use them?

A: Yes. For example, I want to batch export the resumes of qualified students, or batch mark them as "communicated" or "re-interviewed". In addition, I also want to add notes to students, such as "smooth communication and qualified skills", for reference later.

Q: Do you need the system to automatically remind you of unprocessed applications? How often do you hope to be reminded and in what way?

A: Yes. I hope to be reminded once every 24 hours, using system internal messages. If it is not processed for more than 48 hours, send an email reminder again.

Q: Have you ever encountered a situation where the applicant's skills do not match the position during previous recruitments? How do you hope the system will help you solve it?

A: Yes. For example, some positions require data statistics skills, but students cannot even use Excel functions. I hope the system can add a skill test link. When students submit their applications, they must first complete a simple skill test and pass it to enter the screening link. Moreover, the system can automatically generate a skill test report, so I can know at a glance whether the student meets the requirements, which saves a lot of trouble.

MO Interview Result 3

Q: How do you currently post TA position information? About how long does the whole process take?

A: Currently, I post it one by one in class groups and major groups, and also accept offline consultations from students. The whole process takes about 2 hours, mainly spent on forwarding through multiple groups, repeatedly answering the same questions from students, and sorting out consultation records.

Q: What aspects do you mainly focus on when screening TA applicants? What inconveniences do you have when using Excel for screening now?

A: I mainly focus on whether the major is suitable, whether the time is sufficient to cooperate with all working hours of the module, and whether there is patience to answer students' repeated questions carefully. If they have done TA before, work evaluation is better. When using Excel for screening, the conditions are too limited. Fields that cannot be fixed, such as "time flexibility" and "past evaluation", cannot be screened. Moreover, the information filled in by students is not standardized, which easily leads to inaccurate screening results and requires manual verification, which is very troublesome.

Q: What key information do you hope the new system will include when posting positions?

A: I hope it can include these: module leader (that is, my name and contact information), specific responsibilities of TA, specific weekly working hours, multiple optional working periods, multiple optional skill requirements, priority admission conditions (such as scholarship winners, those with TA experience), and when the salary will be paid.

Q: Do you need batch operation or marking functions when viewing and screening applicants' resumes? In what specific situations do you use them?

A: Yes. For example, some students are obviously not qualified, and I want to batch delete their applications; I also want to batch send re-interview notices. In addition, I can mark the advantages and disadvantages of students, so that I can focus on these points during the re-interview later.

Q: Do you need the system to automatically remind you of unprocessed applications? How often do you hope to be reminded and in what way?

A: Yes. I hope to be reminded once at 10 am and 6 pm every day, using system internal messages and WeChat applet reminders, so as to ensure that I process applications in a timely manner without delay.

Q: Have you ever encountered a situation where the applicant's skills do not match the position during previous recruitments? How do you hope the system will help you solve it?

A: Occasionally. For example, some students apply for TA positions in experimental modules, but their practical operation ability is not good. I hope the system can allow me to upload skill description documents related to the position. Before submitting the application, students must read and confirm that they meet the requirements. In addition, the system can automatically calculate a matching score according to the skill information filled in by students, and I can quickly screen out suitable students by looking at the score.

2.1.2 Interview Results of TA Applicants (Students)

Student Interview Result 1

Q: Where do you usually get TA position information now? Have you ever encountered information lag or missed information?

A: I mainly look at notices forwarded by the class study committee, messages in the college teachers' group, and occasionally visit the college official website. There is indeed information lag. For example, some positions have already closed applications when I see the notice, and I miss the registration. In addition, some TA positions of niche modules are only posted in a small range and not synchronized to the class group, so I don't know there is such a position at all.

Q: What materials do you need to prepare when submitting a TA application? What troubles you most when submitting through email, Excel and other methods now?

A: I need to prepare a personal resume, academic transcript, and relevant skill certificates, such as Excel certificates and screenshots of professional software operations. Currently, I mainly send emails to MOs. The most troublesome thing is that emails are easily submerged, and I don't know if the MO has received them. Moreover, every time I apply for a different module, I have to re-edit the email and upload materials, repeating the same thing, which is very cumbersome. In addition, I don't know my application progress and whether I have passed the screening.

Q: What formats do you hope the new system will support when uploading resumes?

A: I hope it can support PDF, Word and JPG formats, and preferably PDF first, because the typesetting of PDF will not be messy. In addition, it would be better if the resume template can be saved. When applying for other positions later, I can directly modify it and use it without re-uploading.

Q: When looking for TA positions, do you need to screen by conditions such as module, workload, and skill requirements? What is the most commonly used screening condition?

A: Yes, I usually screen by module, workload and skill requirements. The most commonly used are module and workload. I prefer to choose modules related to my major, and hope the weekly workload is 5-8 hours, which will not affect normal classes and study.

Q: Do you care about the real-time update of the application status? How do you hope to receive status notifications?

A: I care very much. I hope to receive notifications through both system internal messages and SMS. For example, after submitting the application, I can receive a "application submitted" notification; if I pass the screening, I will be notified of "entering the re-interview"; if I fail, please also tell me, and it is better to explain the reason for the failure, such as skill mismatch, major mismatch, etc.

Q: If the system can provide you with matching suggestions between positions and personal skills, what content do you hope it will include?

A: I hope it can include the matching percentage, the skill points that match me with the position, and the skill points I lack. In addition, it would be better if there are some simple improvement suggestions, such as "need to improve Excel function operation, you can refer to XX tutorial", so that I know how to work hard.

Student Interview Result 2

Q: Where do you usually get TA position information now? Have you ever encountered information lag or missed information?

A: I mainly look at the notice column of the college official website and the WeChat official account, and occasionally hear from classmates. There will be information lag. For example, the official website posts a notice but there is no push reminder, and I forget to check it, so I miss the application time. There are few missed information, but sometimes the position information is not complete, such as not stating the workload and salary, and I have to ask the MO specially, which is a waste of time.

Q: What materials do you need to prepare when submitting a TA application? What troubles you most when submitting through email, Excel and other methods now?

A: I need to prepare a personal resume, self-introduction video, and course score screenshots. Currently, I fill in an Excel form and then send an email. The most troublesome thing is that the Excel form has very strict requirements. If I make a slight mistake, I have to resubmit it. Moreover, after sending the email, there is no receipt, and I don't know if it has been successfully submitted. In addition, different MOs have different requirements for material formats, which need to be adjusted repeatedly, which is very troublesome.

Q: What formats do you hope the new system will support when uploading resumes?

A: I hope it can support PDF, Word and MP4 formats. MP4 is mainly used to upload self-introduction videos. In addition, it would be better if I can preview the uploaded materials to avoid uploading the wrong files.

Q: When looking for TA positions, do you need to screen by conditions such as module, workload, and skill requirements? What is the most commonly used screening condition?

A: Yes, I will screen by module, workload, skill requirements and working period. The most commonly used are skill requirements and workload. I prefer to choose the skills I have for skill requirements, so that the pass rate is higher; I hope the weekly workload does not exceed 10 hours, so that I can balance study and part-time work.

Q: Do you care about the real-time update of the application status? How do you hope to receive status notifications?

A: I care. I hope to receive notifications through WeChat official account push and system internal messages, so that I can check them at any time. In addition, it would be better if I can set a status reminder. As soon as the application status is updated, I will be notified immediately, so that I will not miss the subsequent links.

Q: If the system can provide you with matching suggestions between positions and personal skills, what content do you hope it will include?

A: I hope it can include the matching degree, the matching status of core skills, and position recommendations sorted by matching degree. In addition, it also tells where my good skills can be used in this position, so that I can judge whether I am really suitable for this position.

Student Interview Result 3

Q: Where do you usually get TA position information now? Have you ever encountered information lag or missed information?

A: I mainly hear from classmates or receive direct private message invitations from MOs. There are both information lag and missed information. For example, many positions are not public and only invited privately, so I have no way to know. In addition, some public positions are posted for a short time, and when I see them, the applications have already closed.

Q: What materials do you need to prepare when submitting a TA application? What troubles you most when submitting through email, Excel and other methods now?

A: I need to prepare a personal resume, skill description, and materials that can prove my available working hours. Currently, I either submit paper materials offline or send emails. Submitting offline is very inconvenient, and I have to make a special trip to the college. When sending emails, materials are easy to lose, and I don't know the application progress, so I can only take the initiative to ask the MO, which is very inefficient. In addition, I cannot modify the application after submission. If I fill it in wrong, I can only resubmit it, which is very troublesome.

Q: What formats do you hope the new system will support when uploading resumes?

A: I hope it can support PDF, Word and TXT formats. In addition, it would be better if I can edit the resume online. If I fill it in wrong, I can modify it at any time without re-uploading.

Q: When looking for TA positions, do you need to screen by conditions such as module, workload, and skill requirements? What is the most commonly used screening condition?

A: Yes, I will screen by module, workload, working location and skill requirements. The most commonly used are working location and module. I prefer offline on-campus positions for working location, which is convenient to participate. I choose modules I am familiar with, which makes the work easier and more efficient.

Q: Do you care about the real-time update of the application status? How do you hope to receive status notifications?

A: I care very much. I hope to receive notifications through SMS and system APP push. In addition, I also hope to directly see the application progress in the system, such as "submitted → to be screened → screened → re-interview", so that I can clearly know where I am.

Q: If the system can provide you with matching suggestions between positions and personal skills, what content do you hope it will include?

A: I hope it can include the matching degree, detailed explanation of the lacking skills, and evaluation of the position adaptability, such as "this position is suitable for students who can do XX skills. Your advantage is XX, and what you need to improve is XX", so that I can choose positions reasonably and improve the application pass rate.

2.1.3 Interview Results of Module Organizers (MO)

Administrator Interview Result 1

Q: How do you currently count the TA workload of the whole college? Are there any inefficient aspects in the whole process?

A: Currently, each MO first counts the TA workload of the module they teach, fills in an Excel form and submits it to the college, then I check and summarize them one by one, and finally calculate the total TA workload of the whole college, the workload distribution of each module, and the workload of each student. There are many inefficient aspects: first, MOs do not submit forms in a timely manner, leading to delays in my summary work; second, each person fills in forms in different formats, and I have to adjust them line by line when checking, which is very time-consuming; third, data is easy to be wrong, such as inconsistent workload units filled in by MOs or missing statistics, and I have to communicate and check with them repeatedly; fourth, I cannot know the progress of workload statistics in real time, and can only ask MOs one by one.

Q: What data dimensions do you hope the new system will support when counting workload? Such as module, MO, time period, etc.

A: I hope it can support these dimensions: workload statistics by module, statistics by modules managed by MO, statistics by week, month and semester, statistics of total workload by individual student, and statistics by work type, such as class assistance, homework correction, consultation, etc., which can be counted separately.

Do you care about the balance of TA workload? For example, will some students take on too many tasks? How do you hope the system will help you balance it?

A: I care very much, for fear that students will take on too many tasks and affect their study. I hope the system can set a weekly workload limit for students, such as no more than 12 hours. If the workload applied by students exceeds the limit, the system can automatically remind them and not allow them to submit the application. In addition, the system can automatically count the workload distribution of each student and generate a balance analysis report. If there is an imbalance in workload, the system will automatically remind me, so that I can adjust it in a timely manner.

Q: Do you need hierarchical management for user permissions in the system, such as MO's permission to post positions and students' permission to apply? How to divide them specifically?

A: Hierarchical management is needed. Specifically, as an administrator, I should have the highest authority to view all data of the whole college, modify system settings, and manage the permissions of all users. MOs have medium authority, which can post positions of the modules they teach, screen applicants, count the TA workload of their own modules, but cannot view the positions and applicant information of other MOs. Students only have basic authority, which can view position information, submit applications, view their own application status and workload, but cannot view the information of other students and the statistical data of the whole college.

Q: What requirements do you have for the data security of the system? For example, how often to back up data and how to control access permissions?

A: In terms of data security, I hope to back up data once a day and do a full backup once a month. The backed-up data should be kept for at least 1 year. Access permissions should be strictly controlled. Different roles can only view data within their own permissions and cannot access beyond their authority. There should also be a data encryption function to protect the personal information of students and teachers, such as contact information and grades. In addition, regular data security inspections should be carried out to find and solve security problems in a timely manner.

Administrator Interview Result 2

Q: How do you currently count the TA workload of the whole college? Are there any inefficient aspects in the whole process?

A: Currently, I create a unified Excel form for the whole college, and let each MO fill in and update the TA workload regularly, then I check and summarize it in real time. The main inefficient aspects are: first, MOs do not update in a timely manner, leading to lag in workload statistics and inability to grasp the TA work situation of the whole college in a timely manner; second, multiple people editing the form at the same time are prone to data conflicts and overwrites, leading to data loss or errors; third, if you want to check the workload of a specific dimension, such as the workload of a certain module in a certain period of time, you cannot screen it quickly, and can only find it manually, which is very inefficient; fourth, after the statistics are completed, the statistical report cannot be generated quickly, and manual typesetting and production are required, which is very time-consuming.

Q: What data dimensions do you hope the new system will support when counting workload?

A: I hope it can support these: statistics by module name and module number, statistics by MO's name and department, statistics by day, week, month and semester, statistics by student's name, student ID, major and grade, and statistics by workload calculation standard (such as by hour, by task volume). It is better to be associated with salary, and calculate the salary directly according to the workload.

Q: Do you care about the balance of TA workload? How do you hope the system will help you balance it?

A: I care. I hope the system can set a semester workload limit for students, and at the same time, can see the workload trend of each student. If the workload of a student has been high, the system can automatically remind me and the corresponding MO to adjust it in a timely manner. In addition,

if the system can recommend a suitable workload according to the student's curriculum schedule to avoid conflicts between workload and courses, it will be better.

Q: Do you need hierarchical management for user permissions in the system? How to divide them specifically?

A: Hierarchical management is needed. The administrator's permissions include: managing users (adding, deleting, modifying users), assigning permissions, viewing and exporting data, and modifying system settings. MO's permissions include: publishing and editing positions of their own modules, screening applicants, counting and submitting the workload of their own modules, and viewing relevant data of their own modules. Students' permissions include: querying positions, submitting applications, modifying personal information, and viewing their own application status and workload. In addition, it is better to set temporary permissions. For example, when MOs are too busy, they can temporarily authorize other teachers to help screen applicants, and the authorization will be automatically revoked after the expiration date.

Q: What requirements do you have for the data security of the system?

A: In terms of data security, I hope to do incremental backup every day and full backup every week. The backed-up data should be stored on a dedicated server to prevent loss. Access permissions should be strict. When logging in, use account password plus verification code, and force password change regularly. Sensitive data, such as student grades and complete contact information, are prohibited from being exported. If export is really needed, it must be approved by me. In addition, a data operation log should be established to record the operations of all users. If there is data abnormality, the cause can be quickly found.

Administrator Interview Result 3

Q: How do you currently count the TA workload of the whole college? Are there any inefficient aspects in the whole process?

A: Currently, each MO submits a paper workload statistics form, and I manually enter it into an Excel form, then summarize and check it. The efficiency is very low, and there are many problems: first, the workload of manual entry is large, it is easy to enter wrong, and the verification is very difficult; second, paper statistics forms are easy to lose and damage, leading to data loss; third, the summary speed is slow, and it cannot provide data support for salary payment and work assessment in a timely manner; fourth, it is impossible to monitor the workload of each module in real time, and only after the summary is completed can we know the situation of the whole college.

Q: What data dimensions do you hope the new system will support when counting workload?

A: I hope it can support these: statistics by module type and affiliated major, statistics by the number of modules and workload managed by MO, statistics by semester and academic year, statistics by student's major, grade and workload ranking, and statistics by work completion status, such as completed, uncompleted and over-completed, which can be distinguished.

Q: Do you care about the balance of TA workload? How do you hope the system will help you balance it?

A: I care. I hope the system can automatically balance the workload of each student. For example, if a module is short of TAs, while students in other modules have light workload, the system can automatically remind me to coordinate the deployment of students. In addition, I hope the system can set a workload early warning. If the student's workload is close to the upper limit, it will send a reminder to avoid exceeding the limit.

Q: Do you need hierarchical management for user permissions in the system? How to divide them specifically?

A: Hierarchical management is needed. The administrator should have all permissions to manage all modules, users and data. MOs can only manage the modules they are responsible for, view the applicant, workload and other data of their own modules, and cannot operate other modules. Students can only view information related to themselves, and cannot view the application records and workload of other students. In addition, it is better to grade the administrators, such as main administrator and deputy administrator. The deputy administrator can only help me handle some simple work and cannot modify the core settings of the system.

Q: What requirements do you have for the data security of the system?

A: In terms of data security, I hope to back up data once a day, and the backed-up data should be kept for at least 2 years, and data recovery should be supported to prevent data loss. Access permissions are graded by role. Different roles can only view corresponding data, and sensitive data should be encrypted. Regularly detect and repair security vulnerabilities to prevent data leakage. Unauthorized personnel are prohibited from accessing the system. A login log should be established to record login time, IP address and other information. If there is an abnormal login, it can be checked in a timely manner.

2.2 Analysis of Interview Results

2.2.1 Summary of Core Pain Points

Core Pain Points of Module Organizers (MO)

First, the efficiency of position release is low, forwarding through multiple channels takes a long time, and information transmission is not unified; second, the applicant screening method is cumbersome, Excel screening functions are limited, batch operations and marking are not possible, which is easy to miss and repeat, and data verification efficiency is low; third, lack of skill matching assistance, often encountering situations where the applicant's skills do not match the position, increasing screening costs; fourth, lack of reminder mechanism for application processing, easy to miss important applications; fifth, workload statistics need to be filled in and submitted manually, the format is not unified, and errors are easy to occur.

Core Pain Points of TA Applicants (Students)

First, the acquisition of position information is not timely and there are omissions. Some position information descriptions are incomplete, leading to missing application opportunities; second, the application submission method is cumbersome, requiring repeated editing and uploading of materials, unable to confirm the submission status, and unable to modify the

submitted application; third, there are many restrictions on the format of resume upload, and there is a lack of template reuse function; fourth, unable to view the application status in real time, needing to take the initiative to consult, with low efficiency; fifth, lack of matching guidance between positions and personal skills, applying blindly, with low pass rate.

Core Pain Points of College Administrators

First, the efficiency of counting the TA workload of the whole college is very low, manual verification and summary take a long time, and data errors, conflicts and losses are easy to occur; second, unable to monitor the progress of workload statistics and data of various dimensions in real time, making it difficult to provide decision support quickly; third, it is difficult to control the balance of TA workload, and there is a lack of effective auxiliary tools; fourth, user permission management is not standardized, unable to achieve refined grading, and there are potential data security hazards; fifth, data backup and security protection are insufficient, and sensitive data is easy to leak or lose.

2.2.2 Extraction of Core Requirements

Core Requirements of Module Organizers (MO)

Efficient position release function (including complete core fields and unified release channels); convenient applicant screening function (multi-dimensional batch screening, marking, and note-taking); skill matching assistance function (skill threshold setting, skill test, matching prompt); application processing reminder function (timed reminder, multi-channel notification); convenient workload statistics function (automatic statistics, unified format, quick submission).

Core Requirements of TA Applicants (Students)

Timely and complete channel for obtaining position information (no lag, no omission, complete information); convenient application submission function (materials can be reused, modified, and submission status can be confirmed); flexible resume upload function (supporting multiple formats, preview, and online editing); real-time application status update and notification (multi-channel reminder, can view progress details); matching guidance between positions and personal skills (matching degree, lacking skills, improvement suggestions).

Core Requirements of College Administrators

Efficient workload statistics function (automatic summary, multi-dimensional statistics, quick report generation); TA workload balance assistance function (workload upper limit setting, balance analysis, early warning reminder); refined user permission hierarchical management (permission assignment by role, temporary authorization, permission traceability); perfect data security guarantee (timed backup, data encryption, access control, operation log); real-time data monitoring and query function (can quickly screen and query data of various dimensions).

2.2.3 Common Requirements and Differentiated Requirements

Common requirements: all need convenient and efficient operation processes to reduce manual operations; all need real-time information feedback and reminders; all pay attention to

data accuracy and security; all need the system to have a certain degree of flexibility to adapt to different scenarios.

Differentiated requirements: MOs pay more attention to screening efficiency and skill matching, students pay more attention to application convenience and status feedback, and administrators pay more attention to statistics efficiency, permission management and workload balance.

2.3 Customer Feedback

2.3.1Overall Feedback

All three types of interviewees are dissatisfied with the current TA recruitment and workload statistics methods. They generally believe that the process is cumbersome, inefficient and error-prone, and there is an urgent need for a unified, efficient and convenient TA recruitment system to solve the current pain points. At the same time, they have high expectations for the functions of the new system, hoping that the system can fit the actual work scenario, meet the core needs of each role, improve the overall efficiency of TA recruitment and management, reduce manual operations, and reduce the probability of errors.

2.3.2 Specific Feedback and Suggestions

System Function Level

First, improve the position release and information display function, unify the release channel, ensure timely and complete information, and support multi-dimensional position screening; second, optimize the applicant screening and matching function, add batch operation, marking and note-taking functions, add skill test and matching prompt, and reduce invalid applications; third, build a convenient application submission and status feedback mechanism, support material reuse and modification, and push status notifications through multiple channels; fourth, develop an efficient workload statistics function to realize automatic summary, multi-dimensional statistics, quick report generation, and support workload balance monitoring and early warning; fifth, establish a refined permission hierarchical management system, assign permissions by role, and ensure data security; sixth, strengthen data security guarantee, regularly back up data, encrypt sensitive information, establish operation logs, and prevent data leakage or loss.

System Operation Level

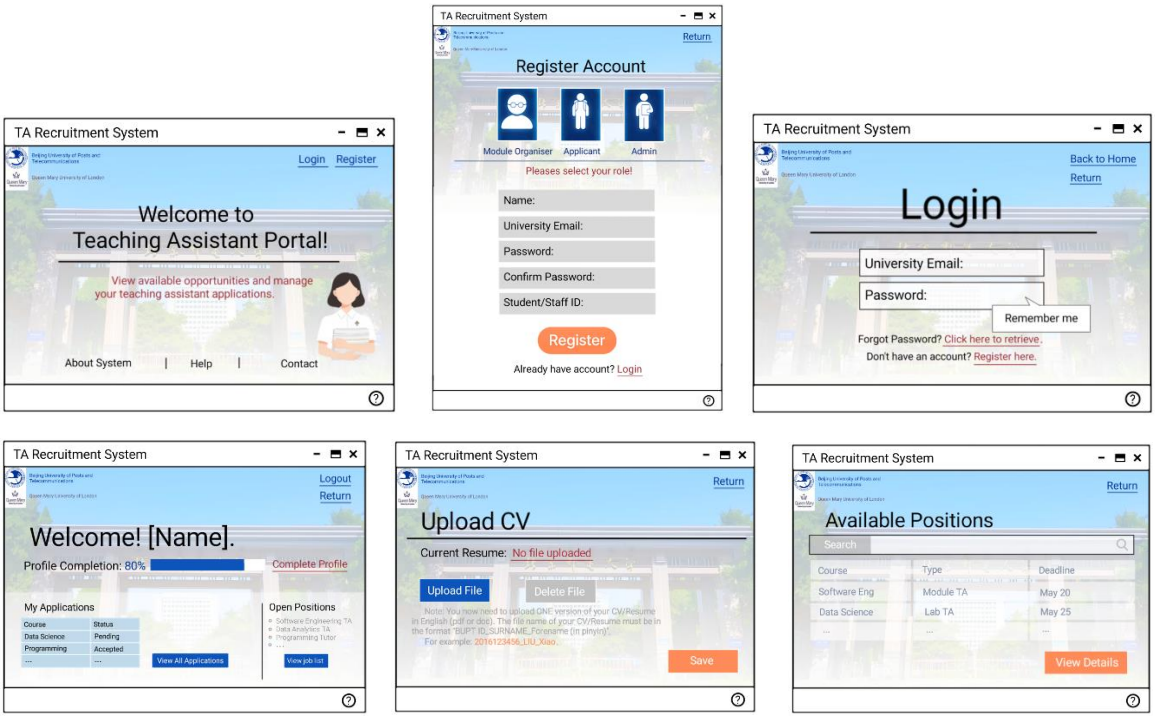
First, the system interface is designed to be simple and intuitive, and the operation process is easy to understand, reducing the learning cost, suitable for users of different ages; second, support multi-terminal login (computer terminal, mobile terminal), facilitating MOs, students and administrators to operate anytime and anywhere; third, provide system use tutorials and frequently asked questions to solve problems encountered by users in the process of use in a timely manner; fourth, support custom setting of some functions (such as reminder frequency, workload upper limit) to adapt to the needs of different colleges and modules; fifth, optimize data import and export functions, support multiple formats, and facilitate data migration and subsequent analysis.

Follow-up Optimization Level

First, after the system is launched, timely collect user feedback, continuously optimize functions, and solve problems encountered in the use process; second, regularly maintain and upgrade the system to ensure stable operation of the system and improve system performance; third, strengthen user training to ensure that MOs, students and administrators can proficiently use all functions of the system; fourth, establish a communication and feedback mechanism, allowing users to put forward functional suggestions at any time to facilitate subsequent iterative optimization of the system.

2.3.3Summary Feedback

Customers (MOs, students, college administrators) unanimously believe that the development of the new TA recruitment system is necessary and urgent, and the core needs are concentrated on the four aspects of "efficiency, convenience, accuracy and security". It is hoped that the new system can effectively solve the current pain points in the process of TA recruitment and management, simplify the operation process, improve work efficiency, ensure data security, and at the same time take into account the differentiated needs of each role, making TA recruitment, screening, workload statistics and other work more standardized and intelligent.

2.4 Prototype design (partial)

Based on the user stories collected and the interface design sketches are as follows: